Harmanpreet Singh Sagar

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EDUCATION

McMaster University Sept. 2022 - June 2027
B.Eng., Software Engineering (Co-op) *GPA: 3.83/4.0*
Awards: Dean's Honour List (3x), Lawrence Tse Research Award (**$6K**), Reilly Scholarship (**$5K**), Award of Excellence
Relevant Courses: Software Design, Data Structures and Algorithms, Databases, Operating Systems, Software Testing

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C++, C, SQL, Bash, Haskell
Frameworks and Libraries: React, Next.js, Node.js, FastAPI, Tailwind CSS, Pydantic, Pytest, JUnit
Developer Tools: Git, GitHub Actions, Jenkins, Docker, PostgreSQL, Vercel, Linux, REST APIs, OAuth 2.0

EXPERIENCE

McMaster University, McSCert Sept. 2026 - Present
Research Assistant, jPipe (Part-time) *Hamilton, ON*
- Traced all **7** failed checks among **187 LLM**-extracted quotes to **5** model errors and **2** PDF parser reading-order faults.
- Built a **Python** text normalizer (**7** ordered rules, e.g. **NFKC**, Markdown stripping) and **5-tier** quote matcher with a **PyMuPDF** fallback, covered by **21 pytest tests**.

McMaster Interdisciplinary Satellite Team Sept. 2023 - Oct. 2026
Firmware Specialist, PRESET CubeSat *Hamilton, ON*
- Owned a **CubeSat Space Protocol (libcsp)** server in **Embedded C** on a **Zynq-7000** (**ARM Cortex-A9**), running as a **FreeAct** active-object task under **FreeRTOS**.
- Ported the CSP stack across **2 board revisions**, reworking **Vitis BSP** configs and CSP port routing.

Develop for Good May 2026 - Aug. 2026
Web Developer *San Francisco, CA (Remote)*
- Delivered **3 of 8** pages of a nonprofit client's **Webflow** site from **Figma** designs over **16 weeks**.
- Recommended **Webflow** over custom code, WordPress, and Wix so a **non-technical client** could maintain it.

Synopsys Inc. May 2024 - Aug. 2025
System Architecture and Modelling Engineering Intern *Mississauga, ON*
- Cut broken builds reaching the **Perforce** depot **90%** by designing a two-tier **MATLAB/Simulink** regression system: a **pre-commit gate** plus a **3-hour nightly** regression.
- Cut the pre-commit gate's runtime **55% (40 to 18 mins)** by parallelizing its **Jenkins CI/CD** pipeline stages.
- Saved **12+ hrs/month** by automating Confluence via the **Atlassian REST API** and **OAuth 2.0** on each passing CI run.
- Drove adoption of the Confluence automation tool to **40+** users across teams after demoing it to **250+** employees.
- Cut **SERDES** debug time **50%** for **6+** colleagues with an **HTTP** trace-observer service in **C++** on **Restinio**.

McMaster University, Department of Computing and Software May 2023 - Aug. 2023
Research Assistant, Drasil *Hamilton, ON*
- Cut functional duplication **25%** in Drasil's **200k+ line Haskell** code generator by refining abbreviation functions.

PROJECTS

F1 Battle Detector | Demo | Code *Python, FastAPI, Next.js, TypeScript, Docker, Vercel*
- Built a **full-stack** F1 race replay (**Python**, **FastAPI**, **asyncio**, **Next.js**) streaming **6** real 2025 races (**167k+** OpenF1 rows) through the live-timing pipeline to rank battles.
- Ran the replay on **stateless Vercel serverless** functions by deriving playback position from the wall clock and rebuilding **108** ticks of state per request in **30-40 ms**.
- Proved replay matches live timing tick for tick over **8 laps** via a fake OpenF1 server, in a **322-test pytest** CI suite.
- Built a **React/TypeScript** dashboard with per-term score breakdowns and a race-long timeline via **SWR** hooks.

SCEMAS | Demo | Code *FastAPI, React, TypeScript, PostgreSQL, MQTT, Docker*
- Led a **5-person team** on a **full-stack** alerting system, owning the **FastAPI** REST facade with **JWT** auth.
- Streamed live alerts to a **React/TypeScript** dashboard over **server-sent events (SSE)** from **MQTT** data.
- Containerized a **3-service Docker Compose** stack (FastAPI, React, **TLS** MQTT) with **4-job** CI and **Trivy** scans.

RBC-BizConnect | Website | Code *Python, JavaScript, SQLite3, Firebase*
- Won **1st place** at **DeltaHacks IX** (RBC) by cutting investor-business matching from **quadratic to linear time**.